Module 4: Securing Access to Cloud Resources

AWS Academy Cloud Developing

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Section 1: Introduction

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Module objectives

At the end of this module, you should be able to do the following:

- Describe the AWS shared responsibility model
- Explain how AWS Identity and Access Management (IAM) helps secure access to AWS resources
- Describe IAM user authentication
- Identify how to authorize access to AWS services by using IAM users, groups, and roles

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Module overview

Sections

1. Introduction
2. Shared responsibility model
3. Introducing AWS Identity and Access Management (IAM)
4. Authenticating with IAM
5. Authorizing with IAM

Demonstrations

- Configuring Cross-Account Access to AWS Resources
- Creating IAM Users and Groups

Activities

Shared Responsibility Model

Knowledge check

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Café business requirement

Frank and Martha liked the proof-of-concept website, and now other team members are going to help Sofía to develop the café‘s web application. Before development continues, she decides to define the level of access that users and systems should have across cloud resources.

```
cursor.close()

sql = "INSERT INTO hive (dev_id, humidity, temperature) VALUES (%s, %s, %s)"

def on_connect(client, userdata, flags, rc):
    print("Connected with result code " + str(rc))

    client.subscribe("+/devices/+/up")

def on_message(client, userdata, msg):
    print(msg.topic + " &&& " + str(msg.payload)+"\n")
    data = json.loads(msg.payload.decode('utf8').replace("'", '"'))
    val = (data["dev_id"], data["payload_fields"]["humidity"], data["payload_fields"]["temperature"])
    connection = mysql.connector.connect(
        host="bees-mariaDB",
        user="bees",
        passwd=mysql_password,
        database="bees"
    )
    cursor = connection.cursor()
    cursor.execute(sql, val)
    connection.commit()
    print(mycursor.rowcount, "record inserted.")
    cursor.close()
    connection.close()

client = mqtt.Client()
```

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Section 2: Shared responsibility model

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Shared responsibility model principles

- **Security** and **compliance** – shared responsibility between AWS and the customer
- **AWS** responsibility:
  - Security **of** the cloud
- **Customer** responsibility:
  - Security **in** the cloud
- Objectives:
  - Reduce the customer's operational burden
  - Provide needed flexibility and control to the customer

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Activity: Shared Responsibility Model

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Activity: Shared Responsibility Model 2

Who is responsible for the security of... ?

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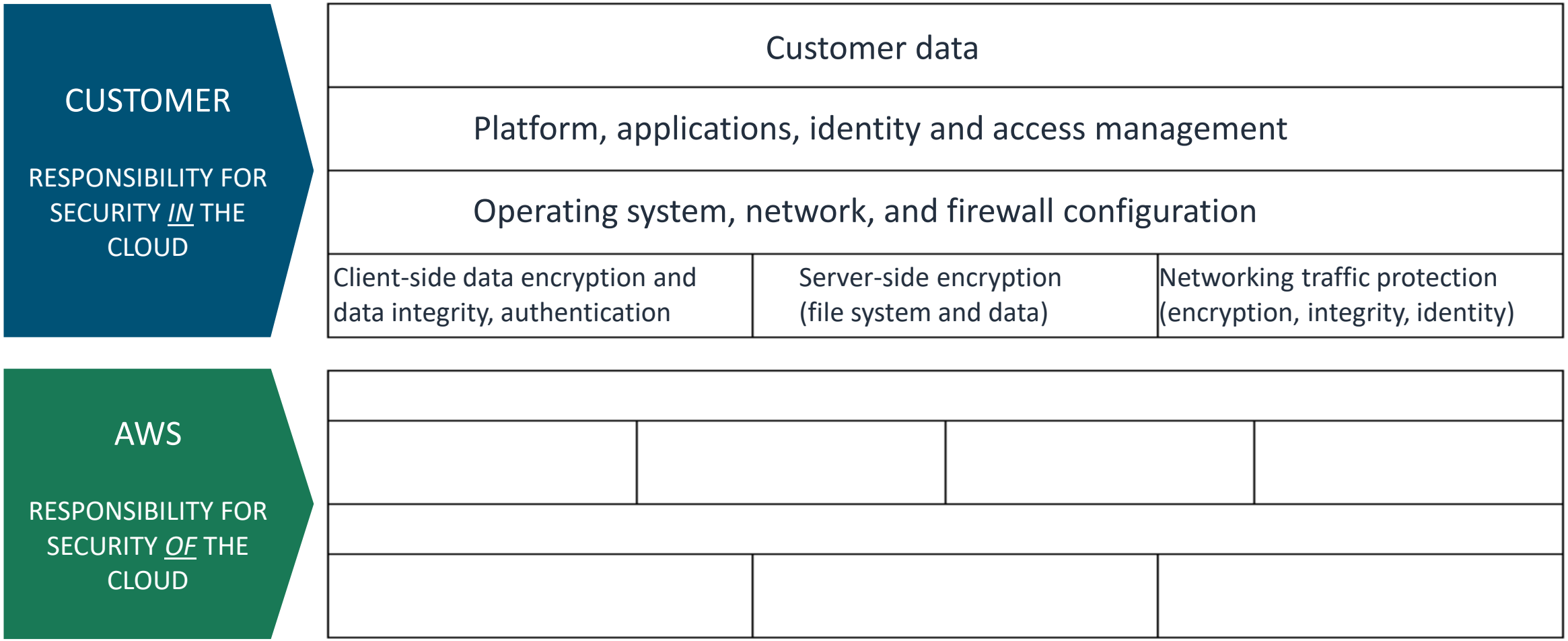
Activity: Shared Responsibility Model

Who is responsible for the security of...



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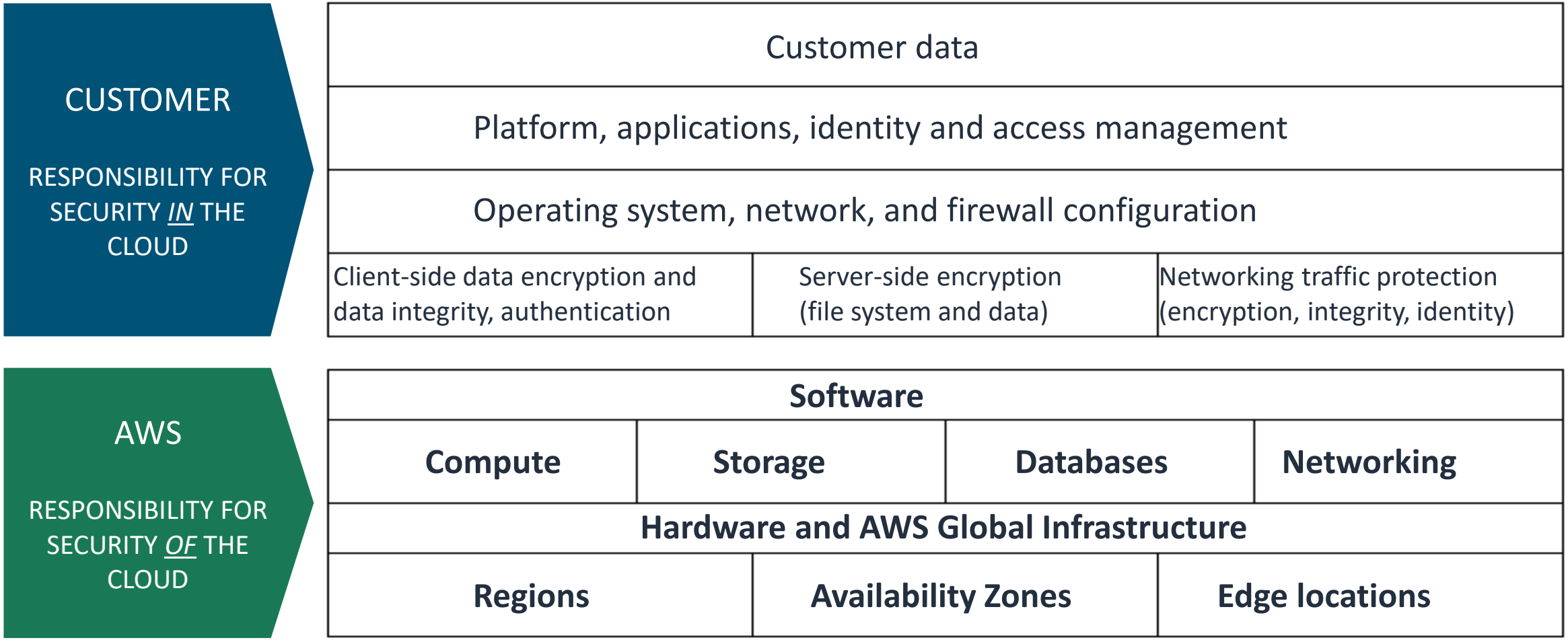
Activity: Shared Responsibility Model

Who is responsible for the security of...



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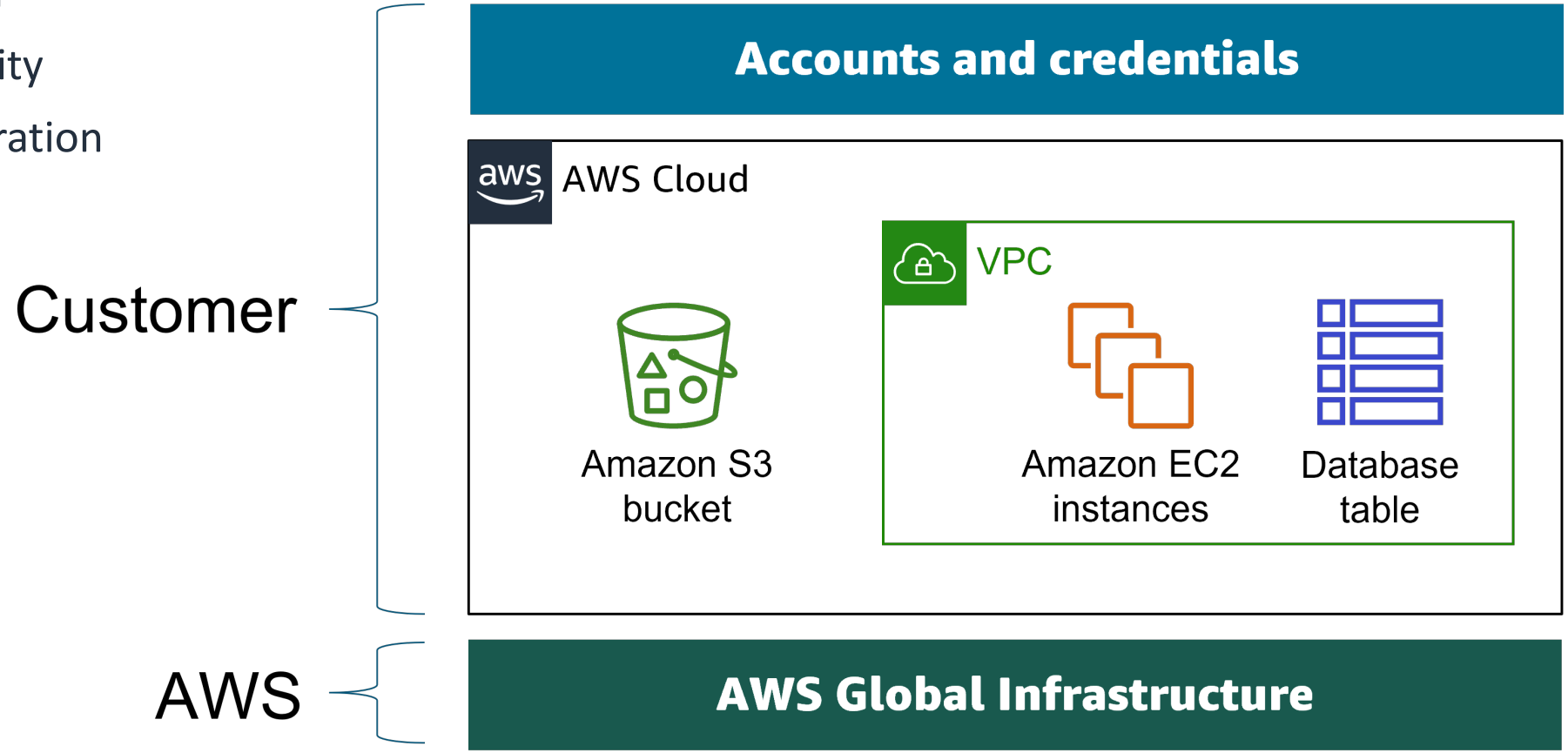
Shared responsibility example

Customer responsibility examples:

- Guest OS configuration
- Application-level security
- Security group configuration



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Section 3: Introducing AWS Identity and Access Management (IAM)

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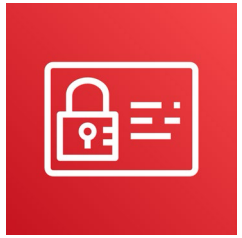
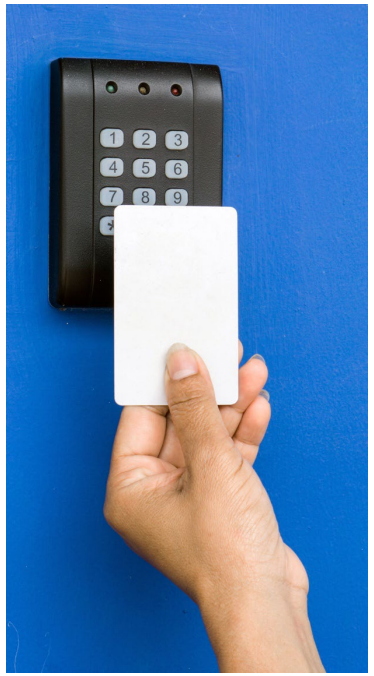
AWS Identity and Access Management (IAM)



AWS Identity and Access Management (IAM)

- Securely controls individual and group access to your AWS resources
- Integrates with other AWS services
- Supports federated identity management
- Supports granular permissions
- Supports multi-factor authentication (MFA)



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Security fundamentals

Authentication

- **Who** is requesting access to the AWS account and the resources in it?
- It's important to establish the identity of the requester through credentials.
- The requester could be a person or an application – IAM calls them principals.

Authorization

- Assuming that the requester has been authenticated, **what** should they be allowed to do?
- IAM checks for policies that are relevant to the request to determine whether to allow or deny the request.

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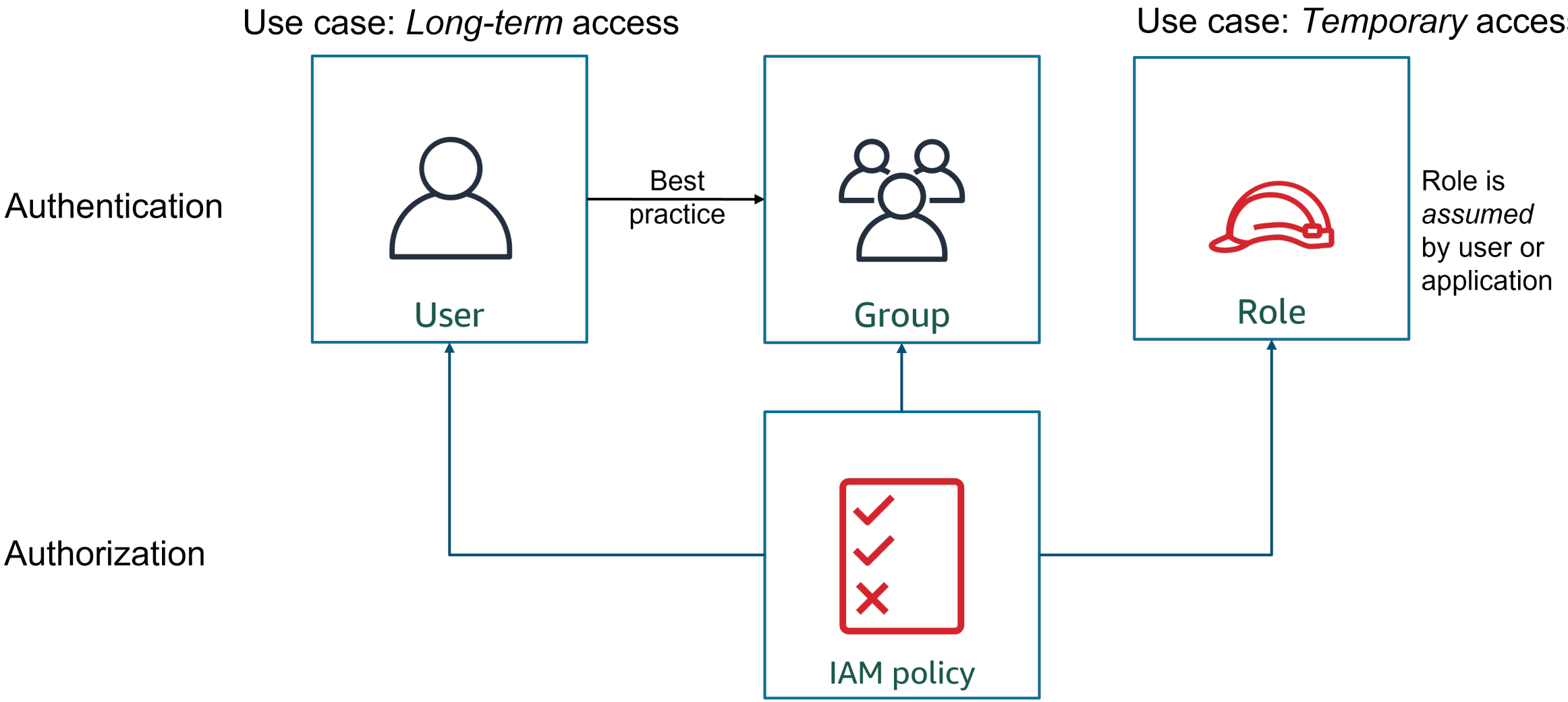
IAM overview



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Section 4: Authenticating with IAM

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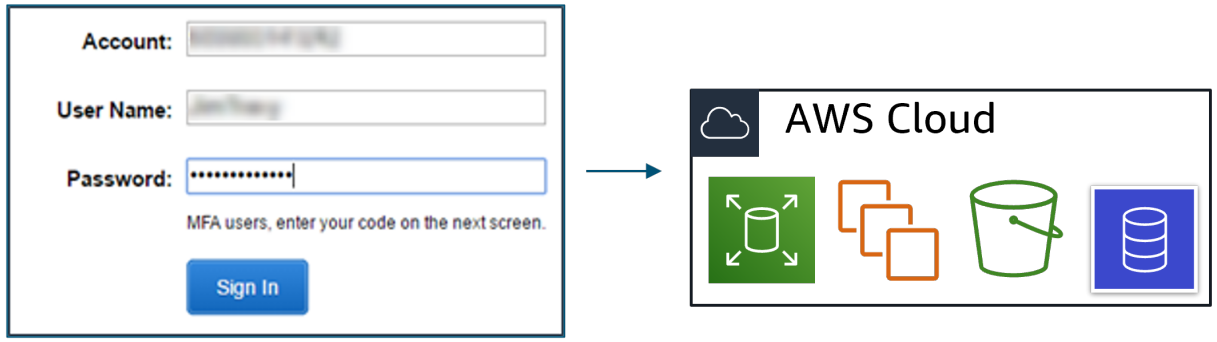
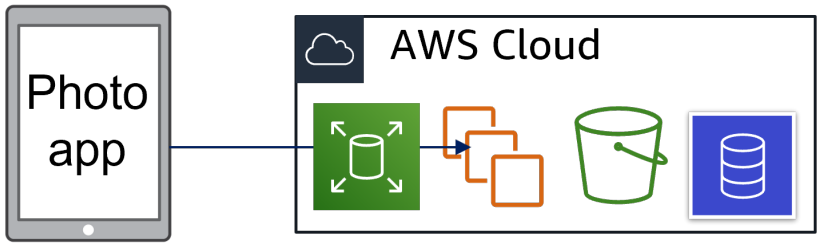
Authentication scenarios

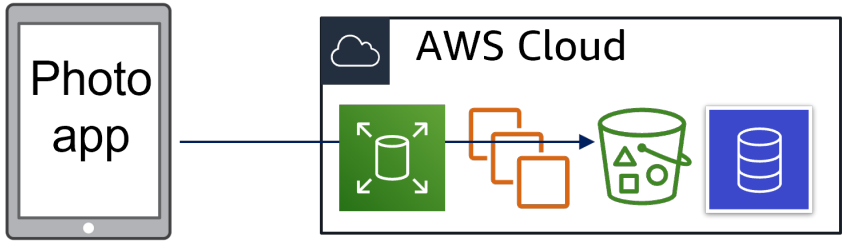
1. AWS authentication
A developer creates a mobile photo app on AWS.



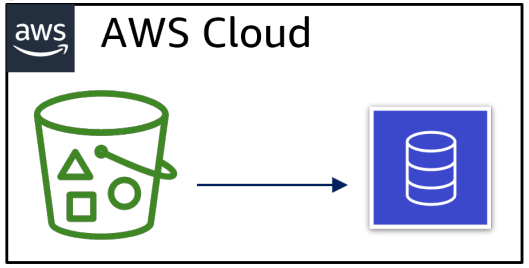
2. Application authentication
A user signs into the photo app.



3. Resource authentication
The user uploads a photo.



4. Database authentication
An Amazon RDS database is updated with an entry for the new photo.



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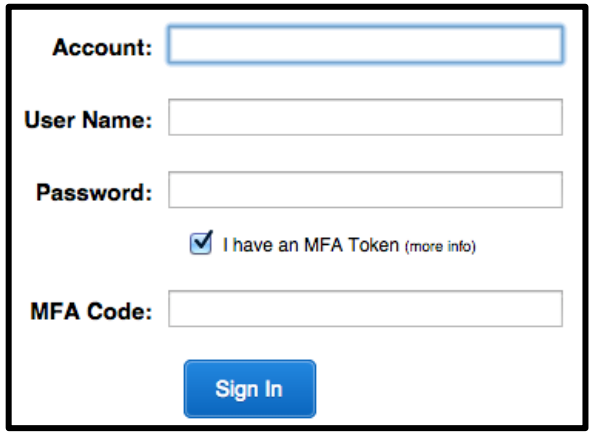
IAM credentials for authentication

User name and password
(console access)



AWS Management Console

MFA Option

Access key ID and secret access key
(programmatic access)

```
[redacted]:~ [redacted]$ aws configure
AWS Access Key ID [****************O22A]:
AWS Secret Access Key [****************4m8i]:
Default region name [ap-southeast-1]:
Default output format [json]:
```

AWS Command Line Interface (AWS CLI)

ACCESS KEY ID
Example: AKIAIOSFODNN7EXAMPLE
SECRET KEY
Example: UtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY

API access

MFA Option

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Multi-factor authentication (MFA)

Adds an extra layer of protection on top of your user name and password

- Users are prompted for an authentication code
- It can be hardware-based or a virtual device

Enable MFA for:

- AWS Management Console users
- AWS API users (requires temporary security credentials)

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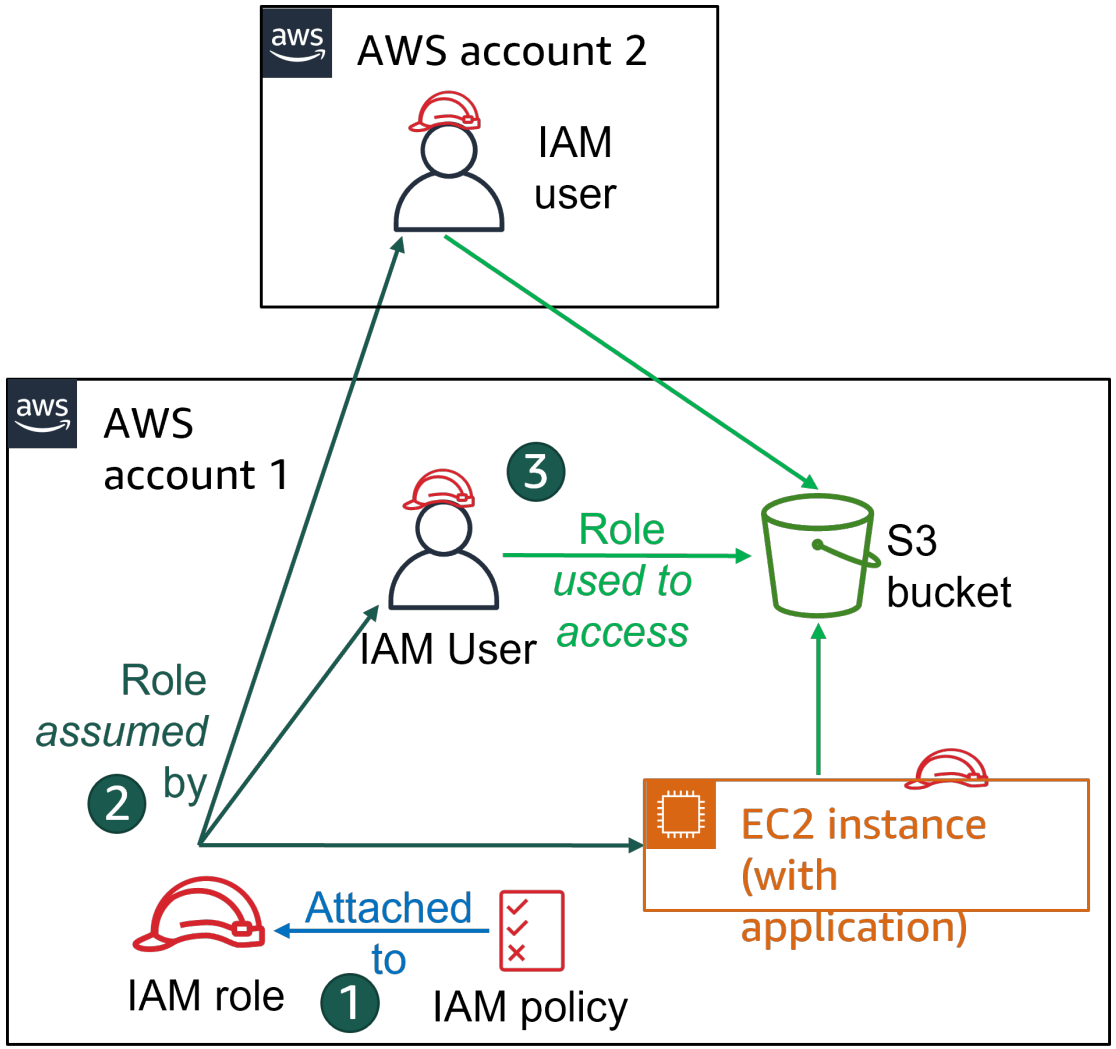
IAM roles provide temporary credentials

IAM role characteristics

- Provides temporary security credentials
- Is not uniquely associated with one person
- Is assumable by a person, application, or service
- Is often used to delegate access

Common use cases

- Application that runs on Amazon Elastic Compute Cloud (Amazon EC2)
- Cross-account access for an IAM user
- Mobile applications



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Demonstration: Configuring Cross-Account Access to AWS Resources

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Providing access to AWS resources through your IDE

- Use AWS toolkits to integrate access to AWS resources.
- Use AWS managed temporary credentials for IDEs installed on an EC2 instance.

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AWS credentials file and credential profiles

Linux, macOS, or Unix: `~/.aws/credentials`

Microsoft Windows: `C:\Users\USERNAME\.aws\credentials`

Example file structure

Profile names

```
[default]
aws_access_key_id = ACCESS_KEY_ID
aws_secret_access_key = SECRET_ACCESS_KEY_ID

[prod]
aws_access_key_id = ACCESS_KEY_ID
aws_secret_access_key = SECRET_ACCESS_KEY_ID
```

```
import boto3
session = boto3.Session(profile_name='prod')
```

Example code—Specifies to use the prod *AWS account*

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Section 4 key takeaways

- **IAM roles** with temporary credentials are preferred for applications that run on Amazon EC2 and for mobile apps.
- Avoid storing AWS credentials in code or in publicly accessible places. **Use credential files** instead.

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Section 5: Authorizing with IAM

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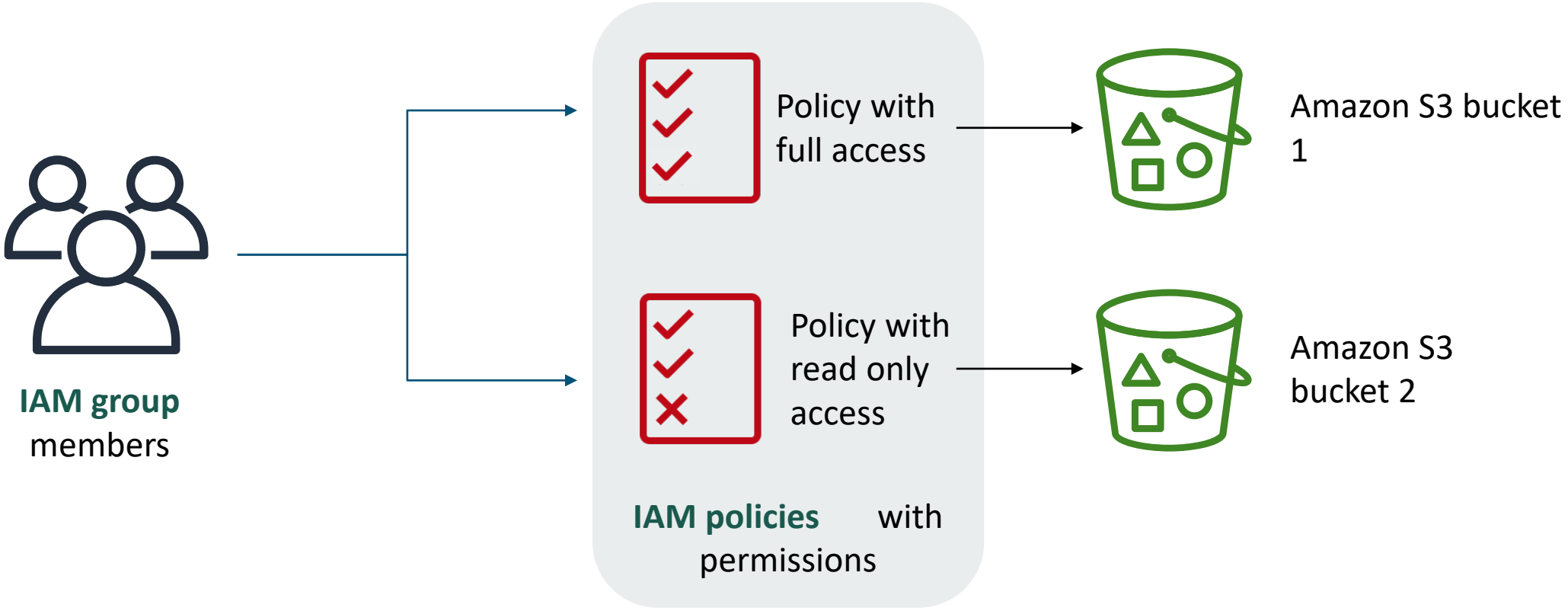
Policies and permissions



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Demonstration: Creating IAM Users and IAM Groups

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Identity-based and resource-based policies

Identity-based policies

What does a particular identity have access to?

User/ group	Has access to	With permission to
Carlos	Resource X	Read, Write, List
Richard	Resource Y	Read
Richard	Resource Z	Read
Managers	Resource X	List
Managers	Resource Y	List
Managers	Resource Z	List

Resource-based policies

Who has access to a particular resource?

Resource	Can be accessed by	With permission to
Resource X	Jie	Read, Write, List
Resource X	Akua	Read, Write, List
Resource X	Mary	Read, List
Resource X	Mateo	List
Resource Y	Paulo	Read, Write, List
Resource Y	Nikki	Read
Resource Y	Mateo	Write, List

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Example of identity-based policy

```
{
  "Version": "2018-10-17",
  "Statement": {
    "Effect": "Allow",
    "Action": [
      "iam:*LoginProfile",
      "iam:*AccessKey*"
      "iam:*SSHPublicKey*"
    ],
    "Resource": "arn:aws:iam::account-
id-without-hyphens:user/${aws:username}"
  }
}
```

The actions allowed

The AWS resources the action can be performed on

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Example cross-account, resource-based policy

```
{
  "Version": "2018-10-17",
  "Statement": {
    "Sid": "AccountBAccess1",
    "Principal": {"AWS": "111122223333"},
    "Effect": "Allow",
    "Action": "s3:*",
    "Resource": [
      "arn:aws:s3:::DOC-EXAMPLE-BUCKET",
      "arn:aws:s3:::DOC-EXAMPLE-BUCKET/*"
    ]
  }
}
```

Who can make the request (Account B)

Actions allowed

Resources shared by Account A

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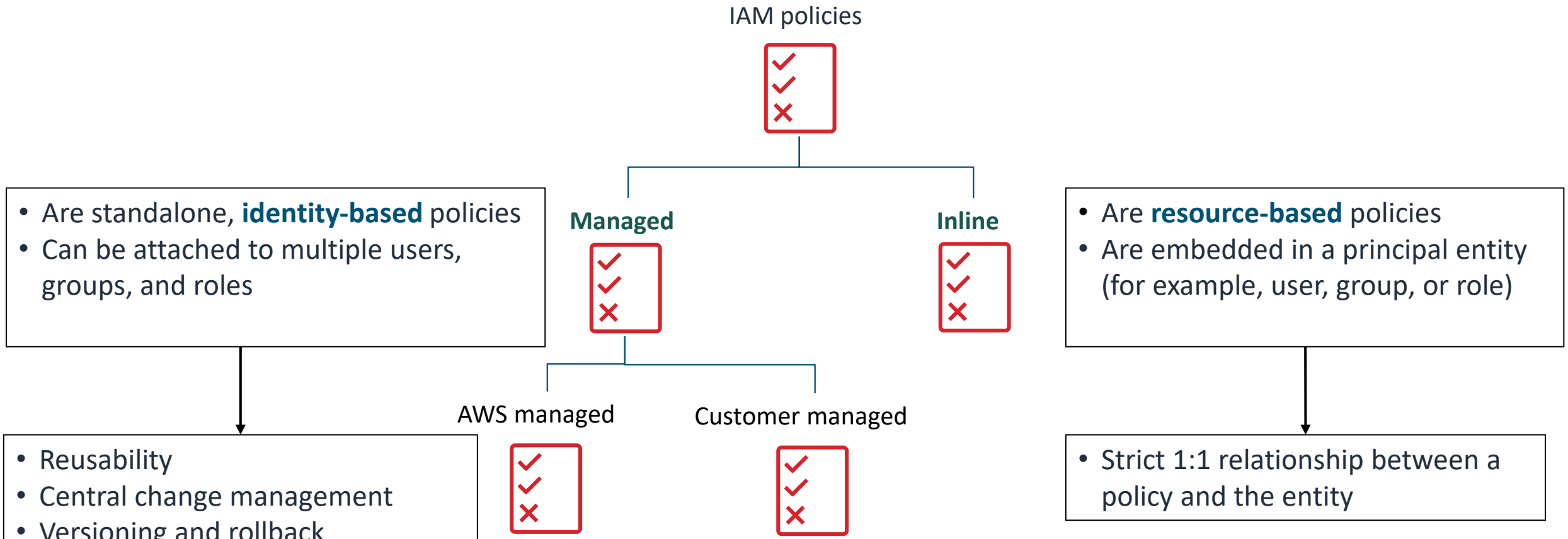
Comparing managed and inline IAM policies



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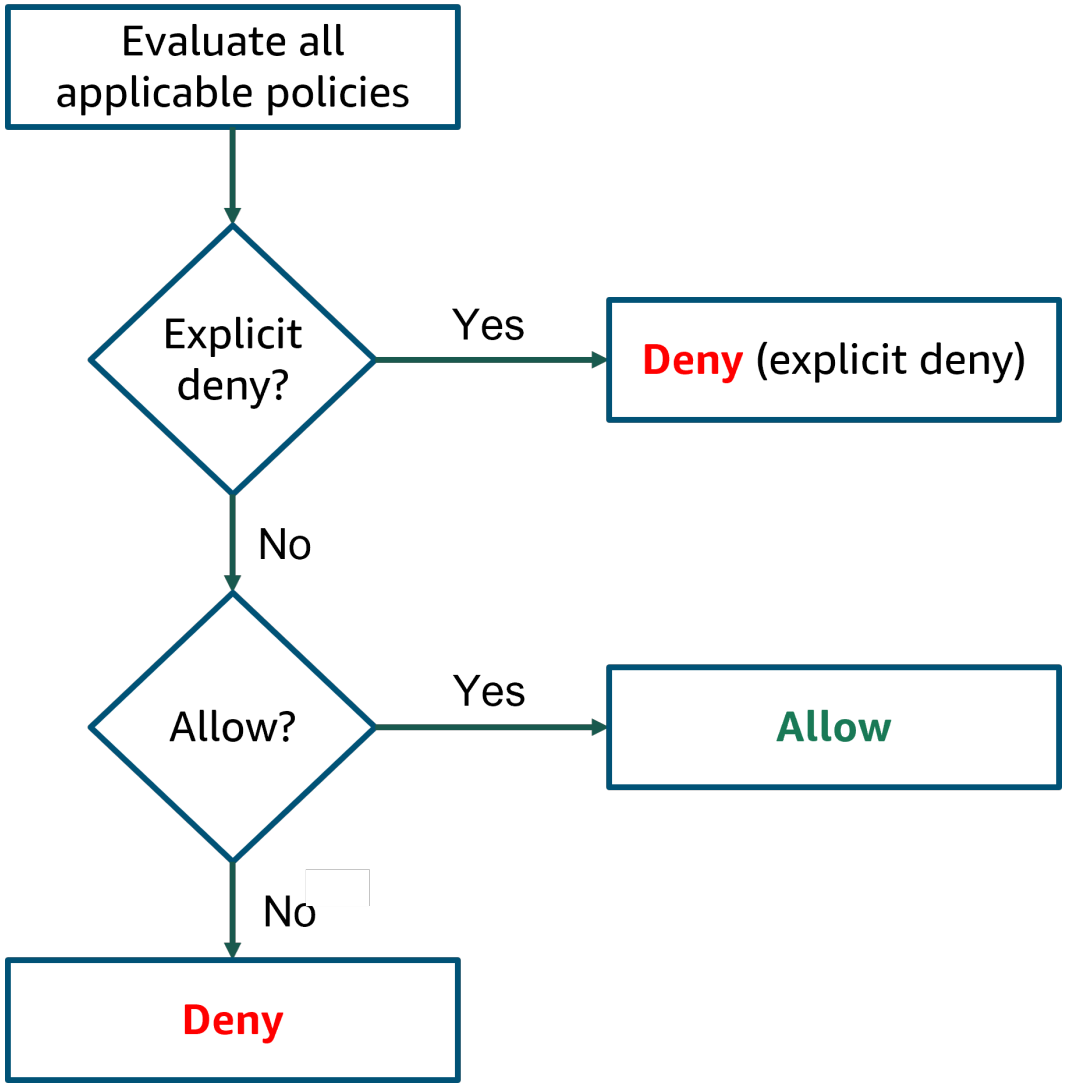
Evaluation logic for IAM policies



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Example IAM policy: Allow statement

Box highlights the Effect Allow, Action, and Resources associated with the Allow statement.

```
{
  "Version": "2012-10-17",
  "Statement":[{
    "Effect":"Allow",
    "Action":["dynamodb:*","s3:*"],
    "Resource":["arn:aws:dynamodb:region:<account-number-without-hyphens>:table/pollynotes",
      "arn:aws:s3:::polly-notes-web",
      "arn:aws:s3:::polly-notes-mp3/*"]
    },
    {
    "Effect":"Deny",
    "Action":["dynamodb:*","s3:*"],
    "NotResource":["arn:aws:dynamodb:region:<account-number-without-
hyphens>:table/pollynotes",
      "arn:aws:s3:::polly-notes-web",
      "arn:aws:s3:::polly-notes-mp3/*"]
    }
  ]
}
```

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Example IAM policy: Deny statement

```
{
  "Version": "2012-10-17",
  "Statement":[{
    "Effect":"Allow",
    "Action":["dynamodb:*","s3:*"],
    "Resource":["arn:aws:dynamodb:region:account-number-without-hyphens:table/pollynotes",
      "arn:aws:s3:::polly-notes-web",
      "arn:aws:s3:::polly-notes-mp3/*"]
    },
    {
    "Effect":"Deny",
    "Action":["dynamodb:*","s3:*"],
    "NotResource":["arn:aws:dynamodb:region:account-number-without-hyphens:table/pollynotes",
      "arn:aws:s3:::polly-notes-web",
      "arn:aws:s3:::polly-notes-mp3/*"]
    }
  ]
}
```

Ensures that the entity can't perform any action on any DynamoDB table or S3 bucket except for the tables and buckets that the policy specifies

An explicit deny statement takes precedence over an allow statement.

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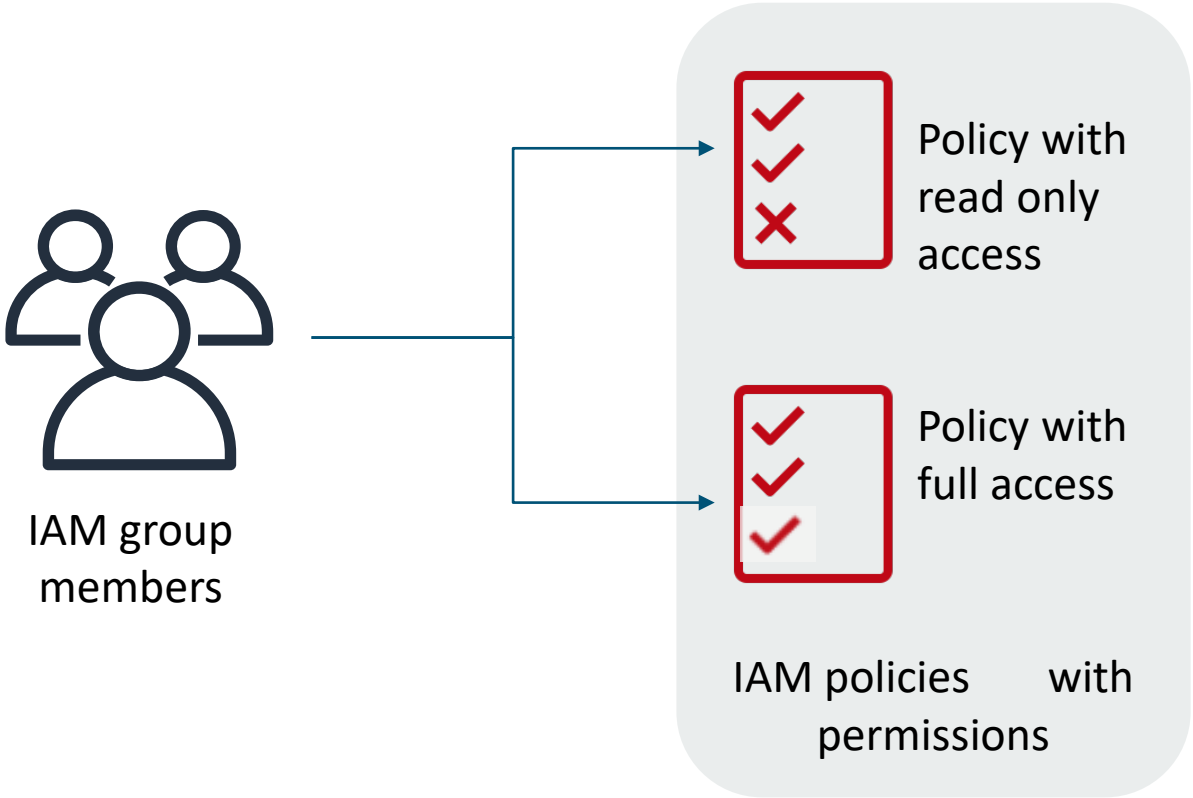
Principle of least privilege



Best practices:

- Start by granting the minimum AWS account permissions that are needed for the job role
- Grant additional access as needed

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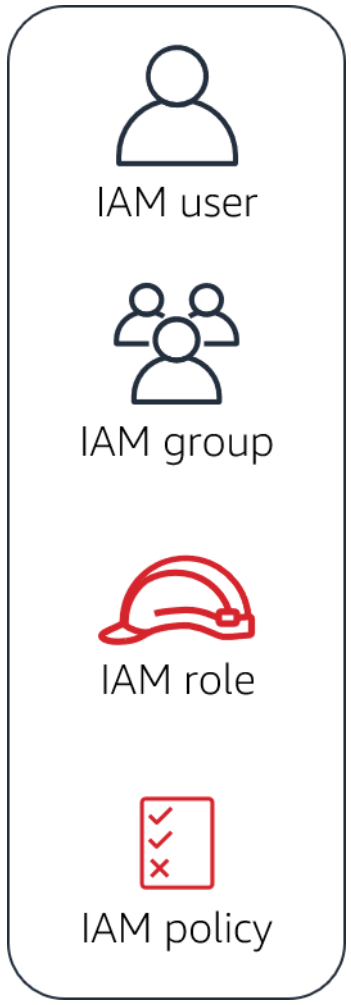
IAM scenario (1)



Account root user

Login: sofia@example.com
Password: *********

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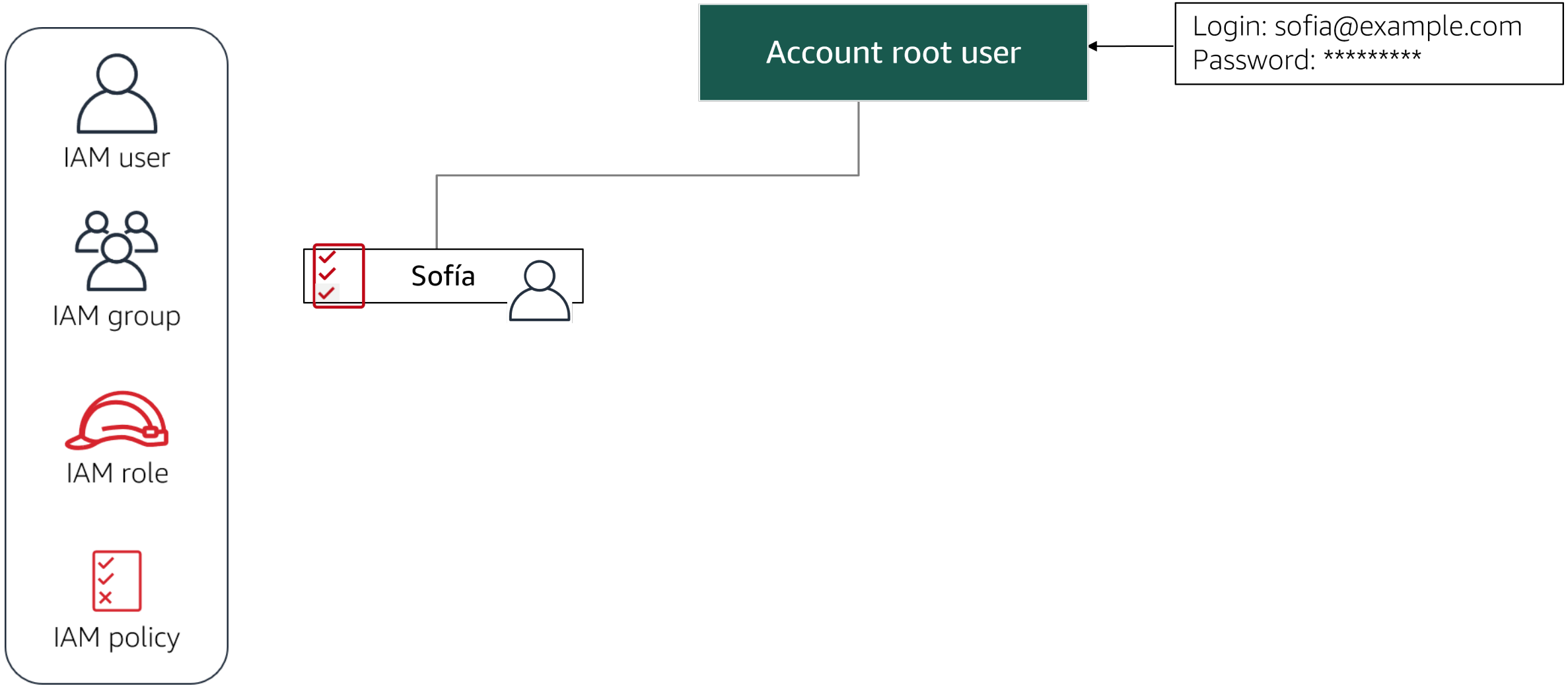
IAM scenario (2)



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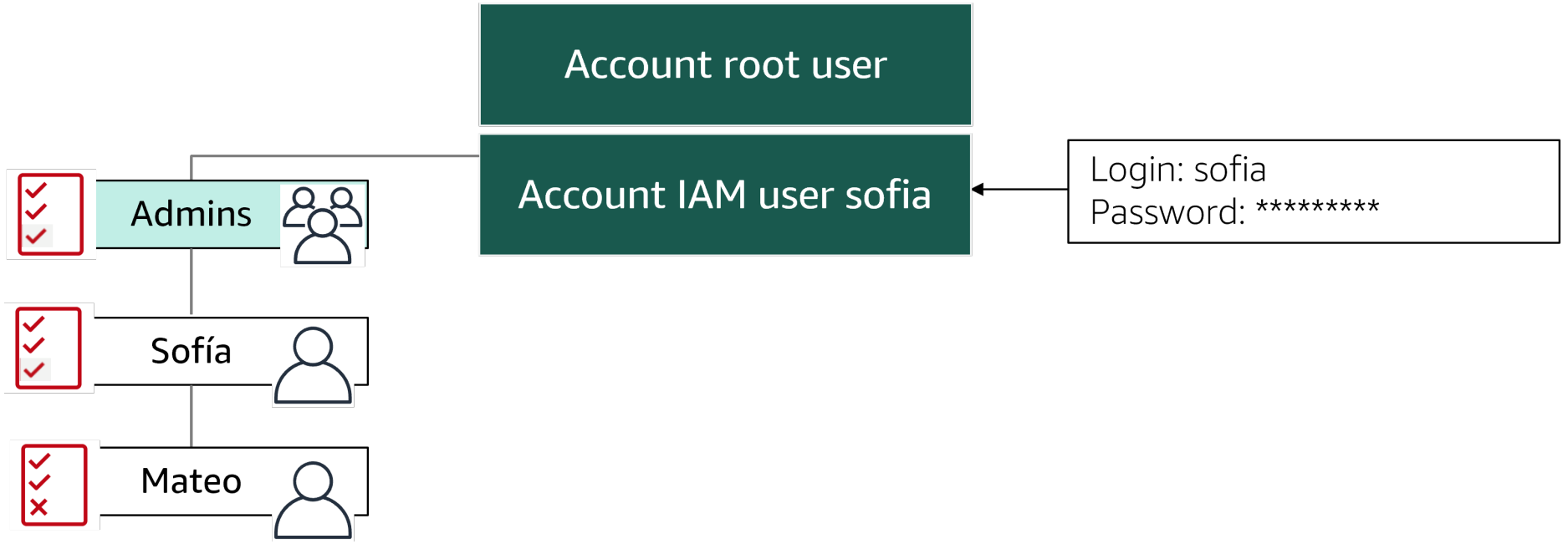
IAM scenario (3)



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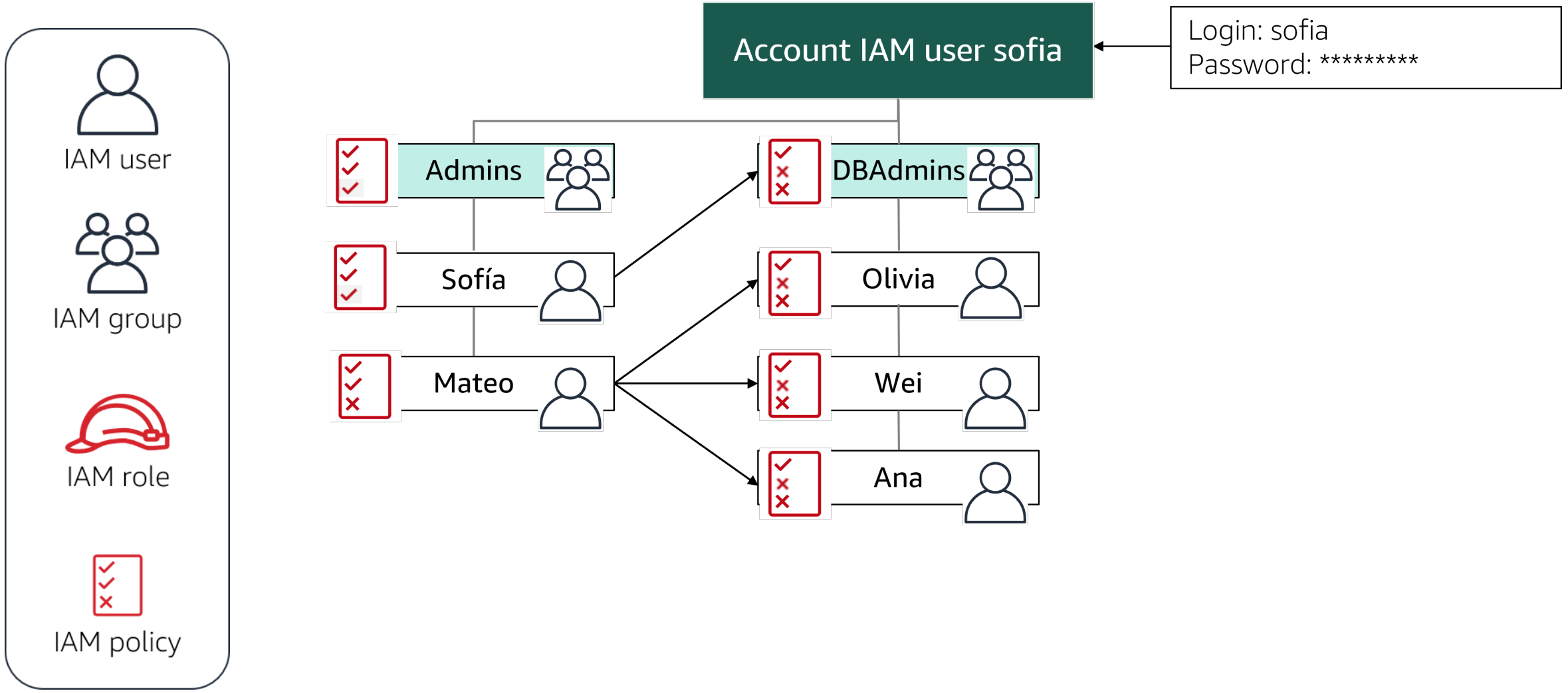
IAM scenario (4)



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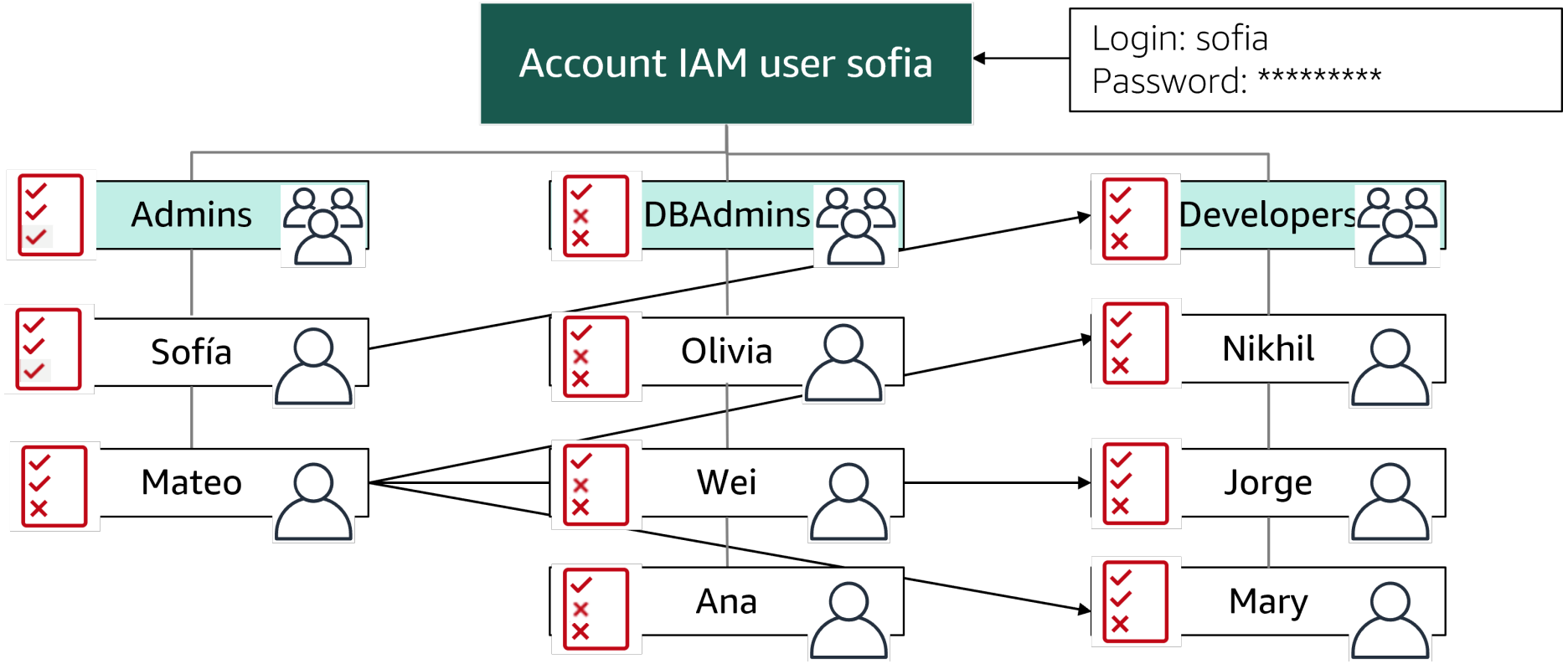
IAM scenario (5)



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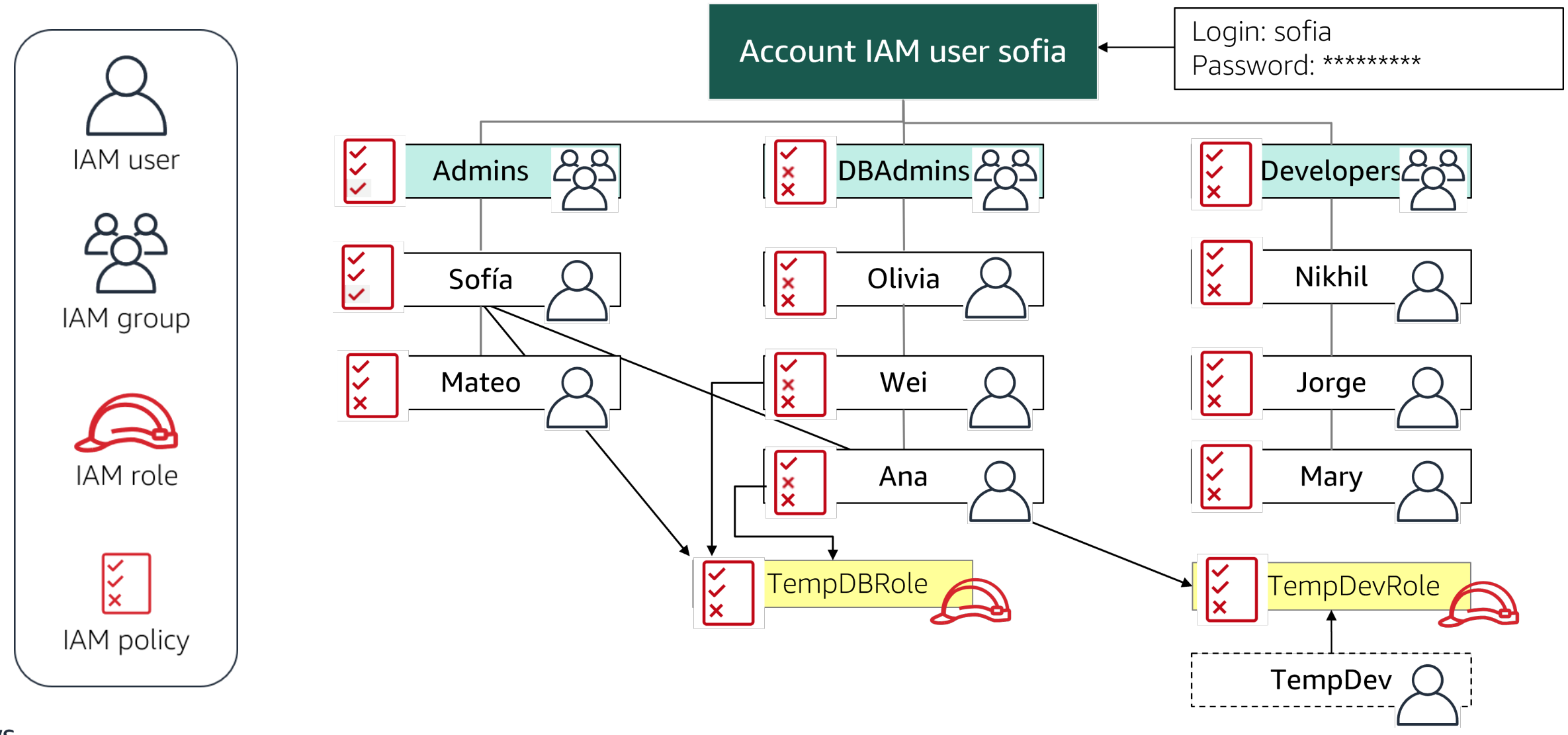
IAM scenario (6)



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Section 5 key takeaways

- Permissions for accessing AWS account services and resources are defined in **IAM policy** documents.
- Attach IAM policies to **IAM users, IAM groups,** or **IAM roles**.
- Follow the **principle of least privilege** when you grant account access.
- When IAM determines permissions, an explicit **deny** will always override any **allow** statement.

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Module wrap-up

Module 4: Securing Access to Cloud Resources

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Module summary

In summary, in this module, you learned how to do the following:

- Describe the AWS shared responsibility model
- Explain how IAM helps secure access to AWS resources
- Describe IAM user authentication
- Identify how to authorize access to AWS services by using IAM users, groups, and roles

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Complete the knowledge check

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Sample exam question

A company is storing an access key (access key ID and secret access key) in a text file on a custom Amazon Machine Image (AMI). The company uses the access key to access DynamoDB tables from instances that are created from the AMI. The security team has mandated a more secure solution.

Which solution will meet the security team's mandate?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **Storing an access key**
- **On a custom Amazon Machine Image (AMI)**
- **To access DynamoDB tables**
- **More secure solution**

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Sample exam question: Responses

A company is **storing an access key** (access key ID and secret access key) in a text file **on a custom Amazon Machine Image (AMI)**. The company uses the access key **to access DynamoDB tables** from instances that are created from the AMI. The security team has mandated a **more secure solution**.

Which solution will meet the security team's mandate?

Choice	Response
A	Put the access key in an S3 bucket, and retrieve the access key on boot from the instance.
B	Pass the access key to the instances through instance user data.
C	Obtain the access key from a key server that is launched in a private subnet.
D	Create an IAM role with permissions to access the table, and launch all instances with the new role.

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Sample exam question: Answer

The correct answer is D.

Choice	Response
A	Put the access key in an S3 bucket, and retrieve the access key on boot from the instance.
B	Pass the access key to the instances through instance user data.
C	Obtain the access key from a key server that is launched in a private subnet.
D	**Create an IAM role with permissions to access the table, and launch all instances with the new role.**

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Additional resources

- IAM FAQs at IAM FAQs
- IAM Tutorials at https://docs.aws.amazon.com/IAM/latest/UserGuide/tutorials.html.
- IAM Policy Reference at https://docs.aws.amazon.com/service-authorization/latest/reference/reference_policies_actions-resources-contextkeys.html.

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